



Unblocking Transparency: Blockchain Readiness and The Philippines' Digital Literacy

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Introduction

The digital revolution has provided further potential for the transformation of governance, offering tools like blockchain technology which may be utilized to enhance transparency in public finances.¹ In the Philippines, where corruption scandals undermine public trust, such as the Commission on Audit's (COA) uncovering of over PHP 42.3 billion in anomalous "ghost projects" from 2023 to 2025, where funds vanished into non-existent infrastructure;² the need for innovative solutions is urgent. Senate Bill No. 1330 (SBN 1330), the "Philippine National Budget Blockchain Act," filed by Senator Bam Aquino on September 2, 2025, proposes a National Budget Blockchain System to digitize budget documents, making them traceable and auditable in real-time.³ This aligns with the Right to Information under Article III, Section 7 of the 1987 Constitution, which guarantees public access to matters of public concern,⁴ as affirmed in *Chavez v. Presidential Commission on Good Government*.⁵

However, the Philippines' low digital literacy rates raise questions about readiness.⁶ With only 40% of Filipinos possessing basic ICT skills according to the Philippine Institute for Development Studies (PIDS),⁷ Many citizens may struggle to utilize blockchain tools like public dashboards. SBN 1330's blockchain system could enhance the Right to Information by enabling public

¹ Celestin Mbonigaba, *The Future of Public Sector Budgeting: How Digital Transformation and Blockchain Are Improving Transparency and Reducing Corruption*, 9 *Brainae Journal of Business, Sciences and Technology* pp. 277-288 (2025), <https://doi.org/10.2139/ssrn.5188103>

² Cristina Baclig, *Flood of Plunder: Billions Paid for Projects Existing Only on Paper*, *Inquirer.net* (September 5, 2025), <https://newsinfo.inquirer.net/2105178/flood-of-plunder-billions-paid-for-projects-existing-only-on-paper>

³ 20th Congress, *SBN 1330 Philippine National Budget Blockchain Act*, Senate of the Philippines (September 2, 2025), https://web.senate.gov.ph/lis/bill_res.aspx?congress=20&q=SBN-1330

⁴ 1987 Philippine Constitution, Art. III, Section 7

⁵ *Chavez v. Presidential Commission on Good Government*, G.R. No. 130716, December 9, 1998

⁶ Rey Mangarin & Jean Climaco, *Exploring Contributing Factors on Poor Digital Literacy of Students: A Review of Existing Studies*, 9(9) *International Journal of Research and Innovation in Applied Science* (October 24, 2024), <https://doi.org/10.51584/IJRIAS.2024.909051>

⁷ Jose Ramon Albert, Francis Mark Quimba, Maureen Ane Rosellon & Mika Muñoz, *Promoting a More Innovative and Inclusive Society Through ICT Development*, Philippine Institute for Development Studies (September 2021), <https://pidswebs.pids.gov.ph/CDN/PUBLICATIONS/pidspn2108.pdf>



access to budget data, but the Philippines' low digital literacy limits its effective use, requiring targeted literacy programs to prevent underutilization and ensure equitable access.

This article examines SBN 1330's potential, blockchain's basics, digital literacy challenges, and overall analysis, arguing that blockchain's value hinges on bridging literacy gaps.

Background of the Study

Budget transparency in the Philippines has been a persistent issue, with the 2023 Open Budget Survey ranking the country 75/100, indicating moderate openness but gaps in real-time oversight.⁸ The current system relies on the Department of Budget and Management's (DBM) Budget and Treasury Management System (BTMS)⁹ and the Philippine Government Electronic Procurement System (PhilGEPS), which provide aggregate data on allocations and contracts but often lack timely updates,¹⁰ allowing anomalies like the COA-flagged ghost flood control projects. For instance, between 2023 and 2025, COA audits revealed PHP 42.3–118.5 billion in economic losses from non-existent infrastructure,¹¹ highlighting the need for better traceability.

SBN 1330 addresses this by mandating a blockchain-based system to record budget transactions immutably, promoting the Right to Information under Article III, Section 7, which empowers citizens to scrutinize public spending.¹² The Senate press release states that budget records would be recorded as digital public assets on blockchain,¹³ to be publicly accessible via a

⁸ International Budget Partnership, *Open Budget Survey* (2023), <https://internationalbudget.org/sites/default/files/country-surveys-pdfs/2023/open-budget-survey-philippines-2023-en.pdf>

⁹ Ruth Abbey Gita-Carlos, *DBM's Digital, Open Governance Promotes Transparency, Efficiency*, Philippine News Agency (December 30, 2024), <https://www.pna.gov.ph/articles/1240583>

¹⁰ Nide Marie Bombay, *Philippines: Case Study on the Philippine Government Electronic Procurement System*, Asian Development Bank (June 2011), <https://www.adb.org/sites/default/files/project-documents/43149-012-reg-tacr-01.pdf>

¹¹ Baclig, *supra* note 2

¹² 20th Congress, *supra* note 3

¹³ 20th Congress, *Bam Aquino Eyes Use of Blockchain Technology to Ensure Transparency, Accountability in National Budget*, Senate of the Philippines (September 2, 2025), https://web.senate.gov.ph/press_release/2025/0902_aquino1.asp



portal.¹⁴ This builds on DBM's July 2025 Polygon pilot.¹⁵ However, with digital literacy at only 40% ICT proficiency,¹⁶ many Filipinos may not engage with the system, risking its full potential.

Blockchain Technology

Blockchain technology, at its core, enables a network of computers to periodically agree on the accurate state of a distributed ledger, ensuring that all participants maintain a consistent and tamper-resistant record of transactions. This decentralized system operates without a central authority, using cryptographic techniques to validate and link data blocks, making it nearly impossible to alter past entries without consensus from the network. As a result, blockchain promotes trust in shared information, with applications in secure record-keeping where integrity is paramount.¹⁷

To better understand this for non-experts like me, imagine blockchain as a communal logbook in a Philippine barangay hall, where every entry (like a budget allocation) is written in ink and bound to the previous one with a lock. Once anything is added, it can't be erased or changed without everyone noticing and agreeing. In the Philippine context, this might look like a nationwide digital "bayanihan" ledger, where government offices and citizens contribute entries on fund releases, and the chain ensures no "ghost" entries slip in, much like citizens verifying community project logs to prevent misuse of shared resources.

In the Philippine context, blockchain can apply to budget transparency by creating a public dashboard where citizens view how funds are allocated, such as in the Department of Budget and Management's (DBM) 2025 pilot, which digitizes release orders and cash allocations for easy online checking.¹⁸

¹⁴ Wilnard Bacionia, *Senate Tackles Blockchain Use to Boost Transparency in Budget Process*, Philippine News Agency (October 2, 2025), <https://www.pna.gov.ph/articles/1260082>

¹⁵ *PH Government Begins Blockchain Use in Budget Release*, Newsbyte.ph (September 30, 2025), <https://newsbytes.ph/2025/09/30/ph-govt-begins-blockchain-use-in-budget-releases/>

¹⁶ Albert et al., *supra* note 7

¹⁷ Suyel Namasudra & Kemel Akkaya, *Introduction to Blockchain Technology*, Springer Nature Singapore Pte Ltd. (2023), https://doi.org/10.1007/978-981-19-8730-4_1

¹⁸ Kenneth Mallari, *DBM's Blockchain Application Officially Goes Live*, Department of Budget and Management (August 29, 2025), <https://www.dbm.gov.ph/index.php?view=article&id=3558:dbm-s-blockchain-application-officially-goes-live&catid=411&highlight=WyJibG9ja2NoYWl1I0=>



For example, the system stores documents on a secure network, allowing users to track budget flows from approval to spending via a web portal.

The benefits include real-time access to information, supporting the Right to Information, and fraud detection, such as spotting ghost projects where funds disappear without results. By making every transaction visible and unchangeable, blockchain could help prevent corruption, as seen in calls for its use in national budgeting to ensure every peso is accounted for.

Digital Literacy: Philippines' Readiness Challenge

Digital literacy in the Philippines remains a significant hurdle, with recent surveys showing that while 93.1% of Filipinos aged five and above have basic literacy, only 70.8% of those aged 10 to 64 are functionally literate,¹⁹ meaning they can read, write, and compute for everyday needs. Meanwhile, the 2024 National ICT Household Survey (NICTHS), conducted by DICT and PSA,²⁰ reveals persistent gaps in technology access and usage, particularly in rural areas.²¹ These challenges extend beyond citizens to government employees and local government units (LGUs), where digital literacy is crucial for implementing technologies like blockchain under SBN 1330.

The connection to blockchain incorporation is clear: SBN 1330 envisions a system where government agencies, departments, and LGUs operate as nodes in a permissioned network, handling data entry, verification, and maintenance. Success hinges on the digital literacy of public sector employees, who must manage these nodes securely. Government workers often lack advanced digital skills, necessitating extensive training to bridge the digital divide.²² The Department of Information and Communications Technology

¹⁹ Philippine Statistics Authority, For Every 10 Filipinos, 9 Have Basic Literacy, While 7 Have Functional Literacy (April 4, 2025),

<https://psa.gov.ph/content/every-10-filipinos-9-have-basic-literacy-while-7-have-functional-literacy>

²⁰ Maria April Cruz, CAR's Digital Landscape Fast Growing, but Faces Challenges in Digital Literacy and Cost-Related Barriers – NICTHS Survey, Philippine Information Agency (October 2, 2025),

<https://pia.gov.ph/news/cars-digital-landscape-fast-growing-but-faces-challenges-in-digital-literacy-and-cost-related-barriers-nicths-survey/>

²¹ Apipa Bagumbaran, New ICT Survey Data Seen as Key to Closing PH Digital Divide, Philippine Information Agency (September 30, 2025),

<https://pia.gov.ph/news/new-ict-survey-data-seen-as-key-to-closing-ph-digital-divide/>

²² Eduardo Andaya, Paving the Digital Path: A Position Paper on Strengthening ICT for Transformative Public Governance in the Philippines, Pamantasan ng Lungsod ng Valenzuela (June 1, 2025),

<http://dx.doi.org/10.2139/ssrn.5335807>



(DICT) has launched initiatives like the CLICK program,²³ which focuses on digital literacy for geographically isolated areas and public sector staff,²⁴ but coverage remains limited in rural LGUs. The government also maintains the PNPKI system as the legal framework for digital signatures,²⁵ and DICT has held training sessions for civil servants on its use.²⁶

Across LGUs, readiness varies. Urban areas like Baguio have piloted blockchain for governance,²⁷ but many rural LGUs are likely to struggle with basic digital infrastructure and skills. A proposed smart city framework for Philippine LGUs notes that while blockchain could enable transformative governance, low digital literacy in local administrations hinders implementation.²⁸ For SBN 1330's nodes, maintenance and security pose additional challenges. The bill would most likely limit the validator nodes to key agencies like DICT, DBM, and COA for security, but nationwide rollout to LGUs requires comprehensive training to prevent vulnerabilities like cyberattacks or errors in data handling. Without adequate literacy, these nodes could face maintenance issues, such as irregular updates or security breaches, undermining the system's reliability.

Analysis

The question we must ask: If the “Philippine National Budget Blockchain Act” were implemented in the Philippines' real-world conditions, would a blockchain-based national budget ledger actually enhance constitutional rights, particularly the Right to Information? Or would it become an impressive

²³ Rainier Allan Ronda, DICT Launches Project CLICK to Boost Digital Literacy, *The Philippine Star* (May 5, 2025), <https://www.philstar.com/the-freeman/cebu-news/2025/05/05/2440643/dict-launches-project-click-boost-digital-literacy>

²⁴ Roel Amazona, 21 Region 8 LGUs Receive Laptops Through DICT's Digital Literacy, *Philippine News Agency* (June 26, 2025), <https://www.pna.gov.ph/articles/1252990>

²⁵ Department of Information and Communications Technology, Philippine National Public Key Infrastructure (PNPKI) (n.d.), <https://dict.gov.ph/pnpki>

²⁶ Department of the Interior and Local Government Region III-Central Luzon, PNPKI Digital Signature Orientation and Training (2022), <https://www.region3.dilg.gov.ph/index.php/field-offices-news/3348-pnpki-digital-signature-orientation-and-training>

²⁷ Artemio Dumlao, Baguio Adopts Blockchain for Transparent Governance, *The Philippine Star* (September 9, 2025), <https://www.philstar.com/nation/2025/09/09/2471443/baguio-adopts-blockchain-transparent-governance>

²⁸ Leovigilda Bithay, Amelia Girly Aranas, Thea Aranas & Kennet Armas, Developing a Smart City Framework for Philippine LGUs: A Policy-Driven Approach to Digital Transformation, 9(7) *Edelweiss Applied Science and Technology* 1704-1719 (2025), <https://doi.org/10.55214/2576-8484.v9i7.8998>



technology that few can use? I argue that blockchain can meaningfully strengthen the Right to Information, but only if the design, rollout, and legal safeguards are deliberately calibrated to the country's institutional and human realities.

1. The Opportunity

At its best, blockchain technology addresses two systemic problems in how government information is produced and shared that weaken the right to information: opacity of transactions and unverifiable records.

A tamper-evident, auditable ledger makes it far harder for funds to disappear into “ghost” projects or for records to be quietly altered after the fact. In constitutional terms, blockchain can make the mechanics of disclosure more effective and efficient; records that are public, persistent, and provably authentic greatly expand meaningful access to information. That is not a mere technical fix; it concretely changes the baseline of what “access” can actually mean in practice—from occasional Freedom of Information (FOI) requests and stale reports to continuous, verifiable visibility into how public money moves.

2. The Practical Constraint: Rights are only as effective as their users

Rights do not exist in a vacuum. The benefit that blockchain delivers is functional: a public that can find, read, and interpret records will enjoy a much stronger Right to Information. But if large portions of the population, or especially the civil servants responsible for maintaining the system, lack the capacity to use the platform, the constitutional promise remains unrealized. In other words, blockchain changes the supply of information, while the constitutional effect depends on the demand side and absorptive capacity. A constitutional right is not fulfilled simply by publishing data; it is fulfilled when the public can meaningfully exercise oversight, challenge discrepancies, and compel remedies.



3. “Effective Transparency”

For SBN 1330 to realize a constitutional improvement, the following are recommended:

Usability and Accessibility. Public portals must present data in plain language, provide contextual explanations, and offer multiple access channels: web, mobile, public kiosks, translated local language, etc. Otherwise, the ledger will be a resource only for experts.

Institutional Routines. Agencies and LGUs must adopt operational practices that integrate the blockchain ledger into ordinary workflows: items must be posted within a fixed timeframe that is strictly followed, agencies must reconcile on-chain and off-chain records as frequently as possible, and the oversight bodies must be able to pull verifiable extracts for audit.

Civic Engagement and Remedies. Transparency yields rights only if citizens and civil society can act on information. The system must include simple grievance and verification channels, whistleblower protection, and clear paths for remedial action when anomalies are found.

Conclusion

SBN 1330 represents a promising step toward leveraging blockchain technology in the Philippines to strengthen the Right to Information under Article III, Section 7, by making budget data more accessible and accountable. The blockchain’s simple ledger system could empower citizens to track public spending in real-time, addressing corruption issues like ghost projects and fostering trust in government operations. However, the Philippines’ digital literacy challenges, evident in gaps among rural populations and public sector employees, underscore a critical readiness barrier. Without adequate skills to engage with blockchain dashboards or maintain secure nodes, the technology risks underutilization, limiting its constitutional impact to only a digitally proficient minority.

However, this does not render blockchain technology useless; rather, it highlights the need for integrated solutions to bridge the divide. By building on initiatives like DICT’s training programs and incorporating user-friendly features



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into the Philippine National Budget Blockchain Act, the government can ensure broader participation, turning potential into reality. Ultimately, effective implementation hinges on prioritizing digital education alongside technological adoption, ensuring that blockchain fulfills its promise of equitable transparency for all Filipinos.

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